



Eye closure sensitivity in juvenile myoclonic epilepsy and its effect on prognosis



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ARTICLE INFO

Article history:

Received 24 December 2012

Received in revised form 12 July 2013

Accepted 16 July 2013

Keywords:

Juvenile myoclonic epilepsy

Eye closure sensitivity

Prognosis

EEG

Theraphy

ABSTRACT

Purpose: To investigate eye closure sensitivity (ECS) in the EEGs of patients diagnosed with juvenile myoclonic epilepsy (JME) and its relationship to prognosis.

Methods: We included 76 JME patients with a minimum follow-up of one year and evaluated a total of 254 EEGs to obtain evidence of ECS. The patients were grouped according to their response to treatment, and these subgroups were compared in relation to ECS and other clinical and EEG features.

Results: There were 12 patients (15.8%) with poor prognosis who showed resistance to appropriate anti-epileptic drug treatment, 15 (19.7%) patients with pseudo-resistance, and 49 (64.5%) patients with good prognosis. The EEGs of only four of the patients displayed pure ECS (5.3%), and only one of these exhibited poor prognosis. Furthermore, 11 patients (14.5%) had both ECS and photosensitivity, and two of these patients exhibited poor prognosis. Thus, neither pure ECS nor ECS with photosensitivity correlated with poor prognosis. A family history of epilepsy and focal findings on the EEG was correlated with poorer prognosis.

Conclusions: ECS is a rare EEG finding in JME and does not appear to be a marker for poor prognosis.

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1. Introduction

Eye closure sensitivity (ECS) is a unique EEG phenomenon that is described as the non-incidental appearance of transient epileptic abnormalities following the closure of the eyes for 1–4 s during a routine EEG recording.¹ The underlying mechanism of ECS has not yet been clarified, and this phenomenon can be observed both in idiopathic focal and generalized epilepsies. Idiopathic photosensitive occipital lobe epilepsy is a typical example of syndromes displaying ECS with focal epilepsy, and focal occipital epileptic activity can be observed just after eye closure in this syndrome.^{2,3} Generalized forms of ECS have also been recorded in most of the genetic/idiopathic generalized epilepsy (IGE) syndromes, such as childhood absence epilepsy (CAE), juvenile absence epilepsy (JAE), and juvenile myoclonic epilepsy (JME), in addition to the proposed syndrome known as eyelid myoclonia with absences (EMA), in

which ECS plays a diagnostic role.^{4–8} It is well-known that EMA has poorer prognosis compared with other absence epilepsies, and ECS appears to be a poor prognostic marker for absence epilepsies.

In contrast, JME is a frequent IGE syndrome that exhibits good response to treatment in the vast majority of patients. However, there is a subgroup of JME patients that exhibit drug resistance.^{9–11} Although a number of the recent studies have investigated the prognostic markers in JME, most of these did not report the effects of ECS.^{12–14}

Our aim was to investigate ECS in JME and its association with prognosis and other clinical features.

2. Methods

We included 76 consecutive patients diagnosed with JME according to the ILAE criteria who had at least one year of follow-up and detailed EEG records available for the re-evaluation of ECS.^{15,16} JME was diagnosed as a specific electroclinical syndrome with bilateral, single or repetitive, arrhythmic, irregular myoclonic jerks predominantly in the arms starting at puberty, and this symptom might be combined with generalized tonic clonic seizures (GTCS) or absence seizures. The EEG should show rapid

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generalized and often irregular spike waves and polyspike waves according to the criteria for ECS.^{15,16}

The data on gender, age at onset of epilepsy, seizure types, febrile seizures, consanguinity, family history of epilepsy, response to anti-epileptic drugs (AEDs), comorbid conditions, MRI findings, and follow-up duration were retrospectively recorded in a standardized form. Although an MRI was not mandatory for the diagnosis, all of the patients had at least one CT.

Patients experiencing more than one GTCS per year and/or more than five myoclonic or absence seizures/clusters per month despite the regular use of appropriately high doses of syndrome-specific AEDs were classified as the “poor prognosis” group (Group I). Patients who exhibited a similar seizure frequency but who irregularly used AEDs or experienced lifestyle problems were grouped into the “pseudo-resistant” group (Group II), and the remaining patients constituted the “good prognosis” group (Group III).¹⁷ The prognosis was evaluated only if the duration of effective therapy lasted more than 12 months.

For the EEG recordings, the electrodes were placed according to the international 10–20 system, and both bipolar and referential montages were used during the evaluations. The patients were instructed to close and open their eyes at least seven times during each EEG recording. Hyperventilation and intermittent photic stimulation (IPS) were also used as standard activation procedures. Our IPS protocol started with the eyes open for 5 s and was followed by the eyes closed for 5 s; stimulus trains of 5–10–15–20–25–30–35 Hz were administered as described elsewhere.¹⁸ All of the available EEG traces were re-evaluated retrospectively by two investigators, and consensus was reached in a joint session when needed.

Generalized spike- or multispike-wave discharges in the resting EEG that appeared on at least two occasions after eye closure (excluding the IPS period) were accepted as ECS. Focal spikes/sharp waves or recurrent 2- to 7-Hz localized slow wave paroxysms were interpreted as a focal abnormality. Spikes and sharp waves representing epileptic abnormalities of variable amplitudes were clearly differentiated from the background rhythms and lasted 20–70 ms and 70–200 ms, respectively. Photoparoxysmal response

(PPR) was diagnosed if the EEG showed localized or generalized spike, multi-spike, or spike-and-wave activities during IPS.

The continuous variables are expressed as the mean $\pm$ SD, whereas the categorical variables are presented as frequencies and percentages. The Chi-square test was used to compare the differences in the categorical variables between the groups. The SPSS 17.0 statistical software was used to perform the statistical analyses. Differences with a *p* value of less than 0.05 were considered statistically significant.

3. Results

We included 24 males and 52 females with a diagnosis of JME, and 49 of the total patients (64.5%) exhibited good prognosis according to our criteria. Table 1 shows the clinical and EEG findings in the study groups.

We noted that Group I included significantly more patients with a family history of epilepsy and focal EEG abnormalities. These focal abnormalities were observed in 40% of the patients in the poor prognosis group (Group I), and these were located mostly in the anterior regions (frontotemporal (3 patients), frontal (2), and central (1)). In contrast, only 4% of the patients in the therapy-responsive group (Group III; frontotemporal (1) and centroparietal (1)) exhibited focal abnormalities.

Of the 72 patients with MRI data, 62 (81.6%) had normal findings. The abnormal MRI findings, which did not differ between the groups, were the following: cerebral atrophy (3 patients), cerebral atrophy and ventricular dilatation (2), pituitary microadenoma (1), encephalomalacia (1), cerebellar hypoplasia (1), meningioma (1), and non-specific hyperintense millimetric lesions (1).

Most of the patients were using valproate (VPA) monotherapy (50 patients) or VPA polytherapy (11) and rest of them were using lamotrigine (4), levetiracetam (5), and topiramate (1) monotherapies and other combinations. However, five of the patients did not use any drugs at the end of the study.

All of the patients had 3 ± 2 (1–10) EEG recordings, and a total of 254 EEGs were re-analyzed for the purposes of this study. The

Table 1
Clinical and EEG features of the study groups.

	Group I <i>n</i> :12 (15.8%)	Group II <i>n</i> :15 (19.7%)	Group III <i>n</i> :49 (64.5%)	<i>P</i> values
Gender (Male/Female)	1/11	6/9	17/32	0.70
Age at onset (year)	15.4 $\pm$ 4 (7–25)	14.1 $\pm$ 2 (12–18)	15.2 $\pm$ 4 (6–31)	0.34
Follow-up duration (year)	9.2 $\pm$ 6 (2–24)	12.1 $\pm$ 10 (1–30)	7.6 $\pm$ 6 (1–23)	0.05
Seizure types				
Myoclonia + GTCS	7 (58.3%)	9 (60%)	36 (73.5%)	0.08
Myoclonia + GTCS + Absences	2 (16.7%)	6 (40%)	9 (18.4%)	
Myoclonia + Absences	2 (16.7%)	–	1 (2%)	
Myoclonia only	1 (8.3%)	–	3 (6.1%)	
Psychiatric comorbidity				
Depression	4 (33.3%)	3 (20%)	5 (10.2%)	0.09
Anxiety disorder	–	1 (6.7%)	1 (2%)	
Somatoform disorders	1 (8.3%)	1 (6.7%)	1 (2%)	
Febrile seizure history	4 (33.3%)	2 (13.3%)	5 (10.2%)	1.00
Family history of epilepsy	6 (50%)	10 (66.7%)	13 (26.5%)	0.005
Consanguinity of parents	2 (17.6%)	1 (6.7%)	13 (26.5%)	0.15
Clinical photosensitivity	3 (25%)	4 (26.7%)	12 (24.5%)	0.55
ECS (+)/PPR(+)	2 (25%)	2 (13.3%)	7 (14.2%)	0.89
ECS (–)/PPR(–)	5 (41.6%)	7 (46.6%)	19 (38.7%)	
ECS (+)/PPR (–)	–	1 (0.6%)	3 (0.6%)	
ECS (–)/PPR(+)	5 (41.6%)	5 (33.3%)	17 (34.6%)	
Focal abnormality	3 (25%)	6 (40%)	2 (4%)	<0.000

Group I: poor prognosis group, Group II: pseudo-resistant group, Group III: good prognosis group, ECS: eye closure sensitivity; GTCS: generalized tonic clonic seizure, PPR: Photoparoxysmal response.

P values reflected the comparison of Group I and III with Fisher exact test.

* *p* < 0.05.

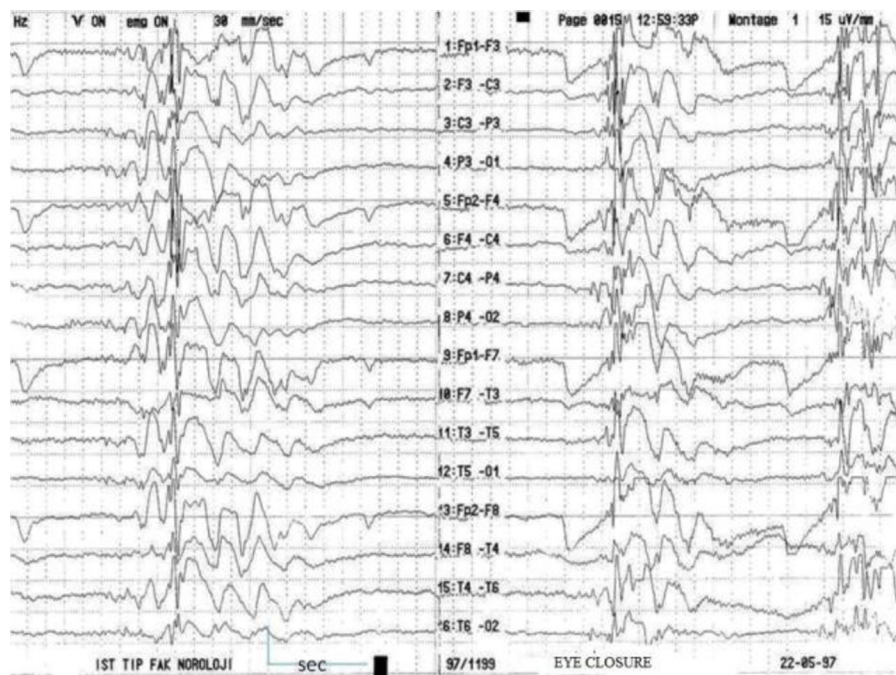


Fig. 1. This EEG example shows generalized multispike-wave discharges in the resting EEG that appear on more than two occasions after eye closure (excluding the IPS period); these discharges were accepted as eye closure sensitivity in this female patient. She suffered myoclonic jerks since 14 years of age. Her seizures are under control with 500 mg/day VPA.

distribution and types of EEG abnormalities, such as ECS and PPR, among the groups is shown in Table 1 and exemplified in Figs. 1 and 2.

There were 12 patients (15.8%) with poor prognosis who showed resistance to appropriate anti-epileptic drug treatments, 15 (19.7%) patients with pseudo-resistance, and 49 (64.5%) patients with good prognosis. The EEGs of only four of the patients displayed pure ECS (5.3%), and only one of these patients had poor prognosis. However, 11 (14.5%) of the patients had both ECS and photosensitivity, and two of these had poor prognosis. Thus, neither pure ECS nor ECS together with photosensitivity showed a correlation with poor prognosis. The 49 patients in Group III were seizure-free, but only 15 of these patients had normal EEGs in the last follow-up (EEG abnormalities persisted in the remaining cases).

Five females of the 15 patients with ECS described marked eyelid blinking during daily life, but no typical seizures of eyelid myoclonia could be recorded through video-EEG examinations. A male patient who did not previously report eyelid myoclonia or blinking was found to have mild eyelid myoclonia in IPS without absences. In contrast, four patients exhibited typical generalized myoclonia in IPS at the time of eye closure. Furthermore, one of our patients reported two GTCs during sunbathing when closing her eyes.

Of the 15 patients with ECS at 19–36 years of age (mean = 27.9), seven had normal EEGs and eight had continuing ECS in their last EEG. Of the 37 patients with photosensitivity at 19–49 years of age (mean = 27.4), 16 still exhibited IPS sensitivity in the last EEG.

4. Discussion

ECS can easily be overlooked when patients are not instructed to close and open their eyes many times during routine EEG recordings. The implications of this rare EEG phenomenon are not known. Brigas et al. reported 11 patients with ECS, of which five had CAE, three had JME, two had JAE, and one had epilepsy with grand mal seizures upon awakening. These findings show that ECS

is not syndrome-specific among the IGEs.⁶ We previously analyzed ECS in a prospective series and found the following result: of 10 patients with ECS, five had JME, two had absence epilepsies, and three had EMA.⁵ In their retrospective analysis of 26 adult patients with ECS, Sevgi et al. found that nine patients had idiopathic occipital lobe epilepsy, one had JAE, four had nonspecific IGE, six had JME, and six had EMA.⁴

EMA, which was first described by Jeavons as a form of photosensitive epilepsy, exhibits distinct clinical and EEG features that justify its recognition as a separate epileptic syndrome according to many reports.^{19–22} EMA can be defined by the following triad: (1) frequent occurrence of eyelid myoclonia with or without brief absences related to generalized epileptic activity and triggered by eye closure, (2) generalized photoparoxysmal EEG response, often in combination with a history of visually induced seizures, and (3) onset in childhood.^{21,22} Seizures occur mainly after eye closures many times a day and are mostly resistant to AEDs, and an eyelid myoclonic status epilepticus is reported in up to one-fifth of patients with this condition. Therefore, eye closure-related discharges appear to be poor prognostic factors.^{20,22}

In contrast, JME is a well-known syndrome with adolescent onset that exhibits predominating myoclonia in addition to short absences and GTCs.²³ It has been well established that most JME patients respond well to AED, but recurrences are common after drug withdrawal.^{23,24} There are some reports of JME patients with drug resistance. The co-existence of all three seizure types is consistently reported to be a risk factor for AED resistance in JME, whereas myoclonus alone or a combination of absence seizures and myoclonus are not.^{9,10,12,25–27} In addition to being an important factor for pseudo-resistance, psychiatric comorbidity is also a cited risk factor for true drug resistance.^{9,10,12} Personality disorders and depression are frequently observed, which emphasizes the need to diagnose these in patients with JME.^{28,29} Our results, which are shown in Table 1, are in line with the abovementioned reports but do not reach statistical significance. It should be remembered, however, that our study design was retrospective and based on a careful re-analysis of available EEGs.

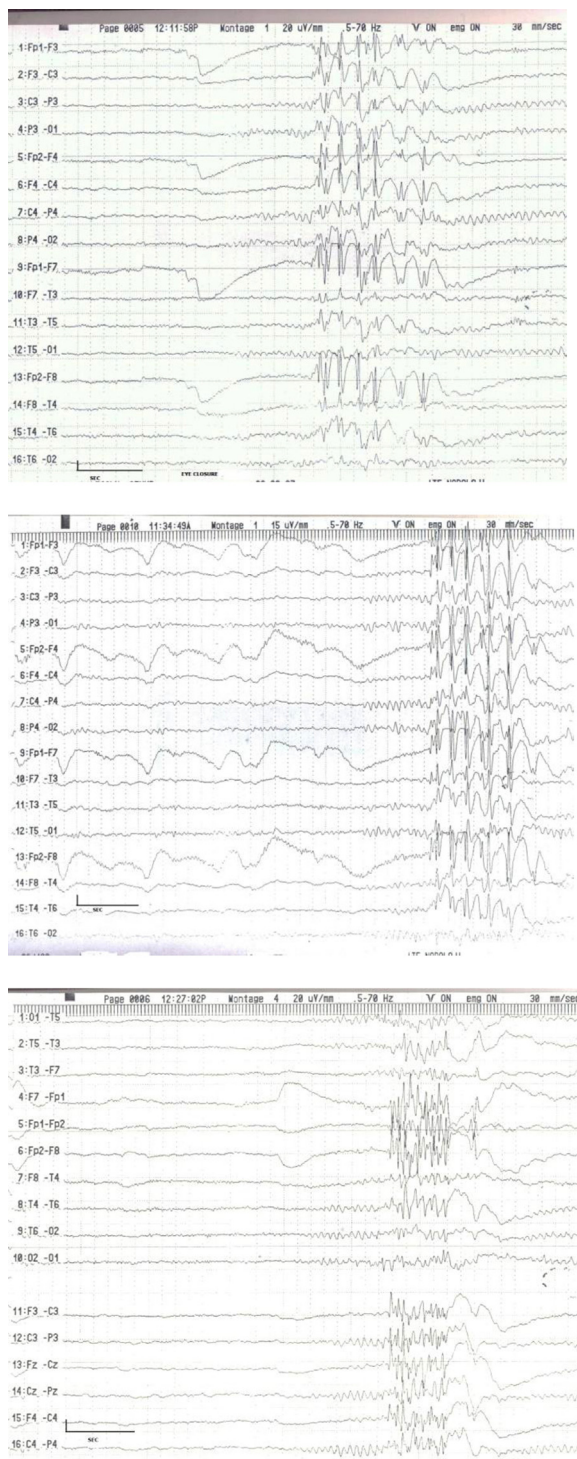


Fig. 2. (a) This EEG shows the eye closure sensitivity of a 30-year-old female patient, (b) an EEG showing generalized spikes and waves in the IPS just after eye closure (same patient) and (c) an EEG showing prominent multispike in the IPS after eye closure (same patient).

Focal EEG abnormalities have been found in 16.5–36.7% of JME patients depending on the methodology.^{12,30} There are reports claiming that focal findings indicate poor response to treatment of JME, whereas other reports did not show any association.^{10–12} A previous prospectively designed follow-up study conducted by our group did not find that focal epileptic findings are markers of poor prognosis.¹² However, our current study included focal recurrent slow waves in the definition of focal abnormality and found an

association with drug resistance. The retrospective nature of the current study may result in a selection bias.

It was also interesting that the current study found that a family history of epilepsy in all relatives or in only first-degree relatives is a marker for drug resistance in JME. Other studies that evaluated family histories of epilepsy did not find family history to be a prognostic marker.^{11,24}

Our results showed MRI abnormalities in 18.4% of the JME patients. Abnormal MRI findings were reported in 12–21% of the JME cases studied.^{31,32}

It has been reported that ECS is present in 15–25% of the patients with JME, which is similar to our results.^{10,33} The relationship between ECS and the prognosis of JME has not been investigated in the relevant literature, although it was reported as an additional finding in two studies. A recent study from Brazil evaluated the endophenotypes of 65 JME patients and suggested that six JME patients with ECS showed significantly poorer responses to treatment.¹⁰ However, it should be noted that their definition of drug response and methodology were different from ours. In another study conducted in Turkey, five JME patients with ECS were found to be seizure-free at the end of the follow-up period.⁴ Thus, it is possible that genetic confounding factors may be responsible for these contradictory results on the prognostic effects of ECS in JME.

It has been suggested that EMA is not an infrequent condition and constitutes 7.5% of the IGEs. In addition, this condition is likely to be under-diagnosed due to its subtle clinical semiology, which can be misinterpreted as tics or mannerisms, and to the masking of EEG changes by the effects of age and AEDs.^{34,35} In some of our patients, ECS was not previously diagnosed if the EEG was not carefully analyzed and did not include repeated eye openings and closings. Video EEG analysis is valuable for the diagnosis or exclusion of eyelid myoclonia and related ECS.

Although these syndromes may show overlap in some cases, JME and EMA could be distinguished based on the predominating type and topography (generalized versus eyelid) of seizures and age at onset.²⁵ It could also be argued that there is an overlap between EMA and JME in the patients with ECS analyzed in this study. However, all of our patients reported the adolescent onset of prominent myoclonic jerks as the predominating seizure type. Reports of eyelid blinking in daily life could not be correlated with ECS in the EEG in almost all of these cases, whereas generalized jerks in the IPS were reported. Therefore, we consider that these patients still should be categorized diagnosed with JME.

Photosensitivity appears at puberty and was reported to disappear at approximately 24 years of age.³⁶ Our retrospective study suggested that both photosensitivity and ECS may persist in the 3rd decade in some JME patients. We were unable to observe any specific drug effect on photosensitivity or ECS in this study.

Electroclinical studies, such as those performed in our study, disclosed prominent but incomplete overlap of photosensitivity with ECS; hence, it has been suggested that the mechanisms underlying ECS and photosensitivity may be different but correlated.³⁴ However, the exact causes are not known. Data on reflex traits suggest that ECS and photosensitivity are expressions of hyperexcitability in JME.¹⁰ The role of the occipital cortex in photosensitive epilepsies is of crucial importance; thus, both eye closures and photosensitivity may synchronize neurons in the occipital cortex depending on many other factors, such as light intensity, volume of cortex activated, level of excitability, and spreading patterns.²¹

5. Conclusions

ECS is observed in approximately 1 out of 5 JME patients. We were unable to show any relationship between ECS and poor prognosis in JME patients. The mechanism of ECS is not known, and

this intriguing EEG finding should be included in future investigations of JME patients to elucidate its unknown implications.

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